Football Match Result Prediction
: based on classification using machine learning

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This study used Kaggle's 'Club Football Match Data (2000–2025)' to build a classification model for predicting football match results and compared and analyzed the performance of various machine and deep learning models. Considering that statistics generated in soccer games are observed only after the end of the game, data were reconstructed with a team-centric to match the actual prediction situation, and previous game-based features were created that summarized information on the past matches of each team. In addition, the time series context was converted into tabular features to improve the limitation that a single time series analysis is difficult due to the multi-team structure. The experiment was conducted with rolling window-based segmentation that maintains time order, and a margin-based threshold strategy was applied to reflect the uncertainty of 'Draw' class. As a result, tree-based ensemble and Boosting-series models performed better than traditional classification models, and CNN was competitive over traditional models but did not outperform boosting-based models due to tabular data characteristics. This study confirmed the effectiveness of the previous game-based feature design and boosting model in predicting the outcome of football matches and suggests the possibility of expansion using time series or graph-based deep learning models and additional contextual information such as players, and tactics.

1 Introduction

The problem of predicting the result of football matches has long been important in various fields such as sports analysis, club management, and the betting industry. In the past, qualitative judgments such as team ranking and the coach's tactical disposition or analysis based on simple statistical indicators were mainly conducted, but these methods have limitations in that it is difficult to consider multiple factors that affect the result of the match at the same time. Recently, as large-scale match record data was accumulated and computational resource is also developed, research is being actively conducted to predict match results based on data using machine learning.

1.1 Background

In sports with a high scoring frequency, such as basketball and baseball, the average effect is relatively large because the number of events that occur during the match is large. On the other hand, football is a low-scoring sport with a low average score per game, and the result of the match can be completely changed with just one-shot success or mistake. So, randomness and variability become larger, and the match result is more probabilistically unstable. In other words, football is difficult to predict the result of the match compared to other sports.

In addition, in football matches, performance indicators such as possession, number of shots, and pass success rate are not necessarily directly proportional to match results. In the content of the match, even though they were dominant, they may lose, and the meaning of statistics may vary depending on the playing style, such as defense-oriented tactics or counterattack strategies. These characteristics make limitations of simple average values or single indicator-based predictions more prominent. Furthermore, various factors such as the player's individual condition, injury, and home advantage are combined into the result of the football match. Many of these factors are difficult to quantify or are often not completely observed with data. Therefore, predicting the result of the football match can be seen as a problem with essentially incomplete information.

1.2 Project Goal

Approaching these problems through machine learning has the advantage of learning complex relationships between various game indicators and allowing models to automatically capture patterns that are difficult for humans to intuitively discover. In particular, in events with low scores and high variability, exactly as the case of football, combinations of several variables are more important than a single indicator, so machine learning techniques that are easy for multivariate analysis will be suitable enough.

With this background, this project seeks to capture the complexity and uncertainty of football based on data, by adding each team's previous match information in addition to the original match statistics and applying machine learning models.

The purpose of this project is to build classification models that predict the result of each team's game as win, draw, and lose with the past match record data. Various models are compared and analyzed to find which approach is suitable for this problem by experimenting with various classification models in consideration of the structural characteristics of football game data.

2 Dataset

In this project, 'Club Football Match Data (2000–2025)' dataset released in Kaggle was used. This dataset is recorded of club football matches played in various leagues around the world from 2000 to 2025 and is data in the form of the large-scale table over a long period of time. Also, this dataset consists of 230,557 rows and 48 columns, and each row expresses one match, and the date of the match, home and away teams, and various statistical indicators and result generated during the match are recorded together.

2.1 Overview

The data includes data on the result of matches, as well as indicators that can indirectly determine propensities of attack and defense. Representatively, information such as the number of shots, shots on target, corner kicks, and fouls of teams is provided, and through this, it is possible to numerically check how many offensive opportunities the team created in the game and how actively they were involved in the defense. In addition, the dataset includes the final score of each game and result information that can distinguish home win, draw, and away win, which is suitable for the classification targeted in this project.

This dataset is proper for building more general match result prediction models that are not dependent on a specific team or league because it covers match records of various leagues and teams. It can also be seen as realistic football match data that naturally includes changes in team power or league environment because it is long-term data accumulated over the years.

In this project, based on this original dataset, input and output variables required for predicting match results were defined, and converted into specific forms suitable for model training in a later process.

2.2 Preprocessing

The key point to predicting results of football matches is not to predict the result of this match using the statistics generated from this game, but to predict the result using only the information already known before the match begins. If variables such as the number of shots, fouls, and corner kicks in this match are used as they are, this is to user the information from after the result has already been determined and does not reflect the actual prediction situation. So, statistics about each match included in dataset used in this project must be shifted in time to a step before the start of the match, and the input features for this match should consist only of information determined before the start of the match and information obtained from previous matches. Regarding the processing of missing values, the uncertainty that may occur in model training is minimized by deleting rows containing all missing values to maintain only complete observations. There are also methods of filling missing values with other values such as the average, but it was determined that there was already enough data in the original dataset, and sufficient data remained for model training even if all rows including missing values were removed, so missing values were not supplemented and dropped.

Most important part of data preprocessing in this project is to reconstruct the data structure into team-centric form. The original data has match-centric structure that includes information of home and away team on one row at the same time, but from the perspective of prediction, what state a team is in at a specific match is more important. Therefore, each game was divided into home team reference data and away team reference data, and all rows were converted to represent one team and one game point. In this process, the column names were unified so that both home and away team had the same variable structure, and a binary variable named HomeAway was added to classify home and away team of the match. This re-organization becomes the basis for easily tracking and combining the past game records of each team in the later stages.

Next, it was considered that the typical single time series analysis method is not suitable, even though football match data has a time series characteristic with an order of time. This is because football match data for this project has multiple teams, with each team competing on a different schedule, unlike the case where a feature is continuously observed over time, such as stock price or temperature data. Additionally, the match always has two teams at the same time, so traditional time-series models, such as ARIMA and a single LSTM, are challenging to apply in this structure. Therefore, instead of directly applying the time series model, this project adopts the method of converting past match information as features and including it in the row of current game. Specifically, for each team, five home games and five away games performed before the corresponding game were extracted, and various summary statistics were calculated from that information. This indicated short-term indicators such as the number of shots, fouls, corner kicks, and team results in the last one game, and medium-term indicators such as the average number of shots, fouls and corner kicks, and accumulated points in the last three and five games. Also, team's Elo and opponent's Elo were also included in the past game standard to reflect the relative level of team power. Since features that compress the time-series context of part one, three, and five games were added to the dataset, the pre-processed data has been in the form that indirectly contains time information while maintaining the tabular structure. In this case, various machine learning classification models such as logistic regression and boosting models can be applied.

Finally, in the process of adding the time series feature, if there is not enough previous game information for specific row, it was removed so that each sample had the same level of past information. Through this process, one refined dataset that can be used for model training for this project was finally constructed.

2.3 Exploratory Data Analysis

For the pre-processed dataset, EDA(Exploratory Data Analysis) was conducted to understand each feature in the dataset more deeply. First, the range, mean, standard deviation, and outliers of features were checked through *describe()* function. And then, after classifying all numerical features and categorical features, distributions of numerical features were visualized as histograms and categorical features were visualized as bar charts to determine the shape of distribution, skewness, and extreme values of each feature.

In the analysis of the target variable, which are results of matches, the frequency and ratio of each class were calculated. Through this, it was found that distribution of classes was not completely balanced, possible risks related to models are tried to prevent in this project even though actually it was not that serious imbalance, and not only accuracy but also macro F1-score were used as evaluation indicators to compensate for this imbalance.

In order to observe how the distribution of results varies depending on the home and away games, the ratio was calculated after grouping based on HomeAway and Target features. And the result shows the phenomenon of home advantage that the winning rate in home games was relatively high.

To confirm the validity of Elo-based features, the difference of Elo between team and opponent in the previous game was calculated. As a result, in the winning game, the Elo difference tends to be skewed in the positive direction, meaning that Elo can be used as the proxy indicator for team power, and positive effects can be expected when the Elo difference is used as a feature for predicting the result of the game.

In a similar way, boxplots for each result were generated for handicaps and game event variables such as shots, corner kicks, and fouls. If the number of shots and corner kicks in the previous game is high, it is more possible to win this game, indicating that the possibility that the attack indicators will provide meaningful information for the prediction.

Also, the linear relationship between the main numerical features and the target was analyzed. Most of features didn't show strong linear relationships with the target, suggesting the possibility that the nonlinear model could be suitable for this problem.

Additionally, ROC-AUC was calculated to evaluate the single-feature baseline classification power of each feature. This is to observe how well a particular feature alone distinguishes the result of matches. The larger value of AUC means more significant, and the value about 0.6 can be judged to be sufficiently significant. Among the features used in this project, it was confirmed that those differences in Elo and recent performance had relatively significant AUC values. Mutual information was also calculated to understand the nonlinear dependence between features and the target. Similar as the result of AUC, some features related to Elo and recent games also showed relatively high values in this case, and a feature of handicaps also showed significant importance.

In the process of checking multicollinearity through VIF(Variance Inflation Factor), there was high possibility of correlation between some variables. Since previously set feature for the Elo difference was calculated as linear computation between other features related to Elo, it is natural that this result comes out. And then, by calculating random forest-based feature importance, features that are importantly used in the tree-based model were identified. This analysis led to the discovery of the possible multicollinearity problem in Elo-related features, and it was decided that the tree-based model, which is robust to multicollinearity, would include all features regarding Elo, while the other model would leave only the feature of Elo difference.

3 Models

In this project, the strategy was adopted to gradually increase the complexity of the model to analyze the prediction of soccer game results, which is a complex classification problem, from various perspectives. First, the reference performance was tested by applying traditional classification models which have high interpretability and computational efficiency, and then the nonlinear relationship learning ability was expanded by using the ensemble-based model and the boosting-based model. Finally, it is possible to explore the possibility of an approach that automatically learns combination of features by experiment with deep learning-based models.

3.1 Traditional Classification Models

In this project, logistic regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and random forest are classified as traditional classification models and applied first. These models have a relatively simple structure but show stable performance for various data distributions and serve as a baseline for predictability of the task intended to be performed.

Logistic regression is a probability-based linear classifier that estimates the probability of belonging to each class through a linear combination of input features. In this project, it was used as the most basic baseline model, and the purpose is to check how much predictive power the previous game-based features have only by linear combination.

K-Nearest Neighbors (KNN) is a distance-based model that finds k number of neighbors most like a specific sample and determines classes through majority voting. Since there is a possibility that games having similar recent game flows, Elo differences, and attack indicators may show similar results in soccer match data, it is expected to be able to utilize these local patterns. In addition, through KNN model, it will be possible to indirectly explore the existence of nonlinear structures in the data.

Support Vector Machine (SVM) is a model that finds a hyperplane that maximizes the margin between classes, and non-linear decision boundaries can be formed through kernel functions. In this project, considering the possibility of interactions between multiple features, SVM is trying to learn more flexible boundaries than simple linear models. It can be expected to bring improved results in that it shows stable performance even in a high-dimensional feature space.

Random forest is a bagging-based ensemble model that randomly generates multiple decision trees and combines the predictions in an average or through voting. Based on tree structure, nonlinear relationships and interactions between variables can be automatically learned, and there is an effect of mitigating the overfitting of individual trees. In this project, it was judged that random forest models could be used as a strong reference model in that various statistical features of the previous games are likely to work complexly.

3.2 Boosting-based Ensemble Models

Boosting-based model is an ensemble model in which the previous stage model sequentially performs learning while giving greater weight to samples that are incorrectly classified. In this project, Gradient Boosting Machine (GBM), Adaptive Boosting, and Histogram-based Gradient Boosting are applied to improve the performance by gradually strengthening the learning for difficult samples.

Gradient Boosting Machine (GBM), is a model that sequentially appends weak learners in the direction of minimizing the loss function. This allows the model to gradually focus on classes with high prediction difficulty, such as 'Draws' in this project, so it can be considered suitable for problems with complex patterns such as predicting football game results.

Adaptive Boosting is a model that conducts learning in a way that increases the weight of misclassified samples. By combining relatively simple learning machines, a powerful classifier can be constructed, and in this project, it can be used as a comparison model to verify the effectiveness of the strategy with boosting.

Histogram-based Gradient Boosting is a gradient boosting deformation model that improves computational efficiency and scalability by discretizing continuous variables based on histograms. Fast learning is possible even on large-scale data and is known to show high performance in tabular data. High performance can be expected in that the data structure of this project is mostly composed of numerical features.

3.3 Deep Learning Models

In this project, the **Convolutional Neural Network (CNN)** is applied as a deep learning-based approach, extending from machine learning-based classification. CNN is a model originally specialized in extracting local patterns from image data, but it can be designed to learn local combination patterns between features by reconstructing input features into a two-dimensional form.

The football match data covered in this project includes several numerical features of different characteristics such as Elo, handicap, and recent game event statistics, and there is a possibility that the combination between them will affect targets. Since the CNN model has the advantage of being able to automatically learn feature combinations that are not directly defined by humans, it is expected that the model will be able to explore patterns from a different perspective from traditional machine learning models.

4 Experiment & Result

Experiments of this project are designed as setting and fitting each classification model and evaluating prediction results using pre-processed data and features based on previous matches. This process proceeded in the order of data splitting that considering time-series characteristics of dataset, margin-based thresholding strategies to solve uncertainty in predicting 'Draw' class, and performance comparison between various models.

4.1 Train-Test Data Splitting

Since the football match data covered in this project has a time series characteristic that is generated in time-order, information leakage may occur in which future information is mixed with past data when random split method is applied. To prevent this problem, rolling window split method was used.

The function implemented for this sets a certain batch size based on the total data length and then divides each batch into 80% of training data and 20% of test data. After that, the start point is moved by a certain step to generate several training-test pairs.

In this experiment, the batch size was set to 50,000, and a total of three divisions were created. In each division, the training data is configured to be in the past section and the test data in the subsequent section to maintain the time flow. Since the results of the logistic regression performed based on the entire dataset without division and the results performed based on the last division were similar, major model comparison experiment is performed based on the last division for efficient experiments. Like the actual prediction situation, this splitting method has a structure that predicts future games based on past game records, making it possible to evaluate the generalization performance of the model in a more realistic way.

4.2 Margin-based Decision Threshold

In the basic multiple classification model, the final prediction is performed by selecting the class with the largest value among the prediction probabilities for each class. However, in football match result prediction, 'Draw' class is often ambiguous compared to other classes, and the simple argmax method may not fully reflect this uncertainty.

To improve this problem, margin-based threshold strategy is used in the experiment. First, for each sample in the model, the difference between the largest probability and the second largest probability is calculated. If this difference is less than the previously set delta, the prediction is judged as low confidence and is forcibly assigned to 'Draw' class. In this experiment, the delta value was set to 0.08, this value serves to induce the model to classify it as 'Draw' if the confidence in winning or loss is not sufficiently great. This method makes the model perceive 'Draw' class as a separate area of uncertainty and enables more realistic decision-making than simple probability maximum-based prediction.

The margin-based strategy constructed in this way has been applied to all previous models, and performance changes can be seen compared to the basic prediction method.

4.3 Performance Comparison

All models were compared fairly using the same train–test division. Performance evaluations were performed based on three metrics: accuracy, macro f1-score, and confusion matrix. Accuracy is the correctly classified percentage of the total predictions, macro F1-score is the value obtained by averaging the F1-score of each class by the same weight, and the confusion matrix is the matrix of classification result between the true and predicted classes.

Traditional classification models showed a certain level of predictive performance, but random forest and boosting models recorded higher accuracy and macro F1-score overall. Boosting-based models, such as Gradient Boosting Machine (GBM), Adaptive Boosting, and Histogram-based Gradient Boosting, showed relatively stable performance even in multi-class environments thanks to their structure of repeatedly correcting difficult samples.

After applying the margin-based thresholding strategy, a tendency to improve the macro F1-score was observed in some models. This can be interpreted because of inducing more active prediction of the draw class.

In addition, a binary classification experiment to predict only for 'Win' or 'Lose' in which the 'Draw' class was removed was also performed. In this case, accuracy increased in most models, suggesting that 'Draw' is a class with high prediction difficulty.

Overall, the boosting-based model showed the best performance, and among them, Gradient Boosting Machine and Histogram-based Gradient Boosting model showed high stability.

In this project, deep learning-based CNN model was tried to perform the task because not all of the models that conducted the experiment did show sufficient performance to see because of classification. Since CNN has the advantage of being able to automatically learn feature combinations unlike traditional machine learning models, it was expected that it would be able to extract potential patterns between previous match-based features by itself. As a result of the experiment, CNN showed more competitive performance than the traditional classification model, but overall, it recorded lower Accuracy and Macro F1-score than the Boosting-based model. This is interpreted because the tree-based ensemble model has a more suitable structure in that this data is not an image or time series raw signal but an already summarized statistical-based table form.

4.4 Discussion

The experimental results show that a model that can effectively learn linear relationships and interactions between variables in the problem of predicting football match results is important. Linear models such as logistic regression can capture basic trends, but do not fully utilize the complex patterns created by previous match-based features.

On the other hand, tree-based ensembles and boosting models showed better performance by being able to automatically learn these nonlinearities and interactions. In addition, margin-based thresholding strategy is thought to have contributed to dealing with high-uncertainty results more realistically, such as 'Draw' classes.

Multiple classification problems including mediate target, such as 'Draw', are still more difficult than the binary classification, which can be interpreted as being due to the high volatility and low scoring structure of the football match itself. These characteristics are expected to be further improved through a model including player-level data, tactical information, or longer time series context in the future.

In addition, CNN's relatively lower performance than the boosting-based model is related to the fact that the input data of this project is already a statistically summarized set of features. CNN originally has strengths in spatial or sequential data where local patterns clearly exist, but it is difficult to fully demonstrate these strengths in current tabular data. Furthermore, this can be seen as a result showing the importance of model suitability according to the data type in prediction problems.

5 Conclusion

Using various club football match records and previous match-based features, this study predicted game results through machine learning models, compared and analyzed various models. By presenting a pre-processing strategy that considers the time series characteristics of soccer game data and a multi-team structure, and a margin-based decision-making strategy to alleviate the uncertainty of predicting a draw, we tried to reflect a realistic prediction environment.

5.1 Prediction Performance

As a result of the experiment, the traditional classification model showed basic predictability, but overall, tree-based ensemble models and boosting-based models recorded higher accuracy and macro F1-score. Boosting-based models showed suitable performance for predicting football match results by effectively learning nonlinear relationships and interactions between features.

In addition, when a margin-based thresholding strategy was applied, the macro F1-score tended to improve in some models, which is interpreted because of inducing the model to make more careful judgments about classes with ambiguous boundaries such as ‘Draw’. In the binary classification experiment between ‘Win’ and ‘Lose’, which removed ‘Draw’ class, the overall accuracy was increased, suggesting that ‘Draw’ is essentially a class with high prediction difficulty.

On the other hand, CNN-based deep learning models did not outperform boosting-based models in this experiment, despite having more complex expression capabilities than traditional machine learning models. This suggests that the data structure of this project is more suitable structure for tree-based models and shows that deep learning models do not always guarantee good performance.

As a conclusion, this project shows that the combination of previous match-based features and boosting-series models is effective in predicting football match results.

5.2 Limitations

There are several limitations to this project. First, the data are limited to team-level statistics and detailed factors such as individual player's condition, injury information, lineup change, and tactical information are not included. These factors have an important influence on the actual result of the match but were not reflected in this project. Second, since past match information was summarized and used in a fixed window size, there is a possibility that the long-term trend or seasonal flow will not be sufficiently reflected.

Third, ‘Draw’ class still shows lower predictive performance than other classes, which is a fundamental problem resulting from the low scoring structure and high variability of football matches. Finally, in this study, experiments have been conducted mainly on traditional machine learning models and tree-based ensemble models, and more complex structures such as time series deep learning models and graph-based models have not been sufficiently explored.

5.3 Improvements & Extensions

In future studies, prediction performance can be improved by including additional contextual information such as player-level data, injury and participation, tactical indicators, and weather information. In addition, it is a meaningful direction to expand Elo-based features in the form of a trend over time as well as a single viewpoint difference.

On the model side, a method of directly inputting the previous game record as sequential data can be considered using time series deep learning. Furthermore, there is a possibility of expanding the Graph Neural Network (GNN)-based approach that expresses the relationship between teams in a graph structure.

From a deep learning perspective, performance improvements are expected if CNNs are extended to a structure where previous match records are entered in time-based sequence form and combined with other time-series methods, rather than a simple two-dimensional feature input scheme. This method may develop the temporal pattern to be directly learned by explicitly transferring the order of the previous matches to the model.

5.4 Implications of the Study

This project shows that machine learning can be used as a practical analysis tool in the problem of predicting football match results. Instead of using the original match statistics as they are, the approach of designing previous match-based features in consideration of the reality of the prediction point and converting time series context into tabular data is also applicable to various sports analysis problems.

Academically, it is significant in that it systematically compared the performance differences by model type in complex sports data. In practice, it is possible to establish a prediction system based on the framework of this study in club analysis teams, sports data companies, and fan participation prediction services.

Taken together, this study couldn’t derive models with enough performanc, but it is meaningful that it simultaneously presents the possibility and limitations of data-driven approaches for predicting football match results and provides a foundation for expanding into more detailed sports analysis systems in the future.